

Date: 27 July 2026

TO WHOM IT MAY CONCERN

This is to certify that **Md. Mahfuz Haque**, a student of the 4th Year, 2nd Semester, Department of Local Government and Urban Development, Jatiya Kabi Kazi Nazrul Islam University (JKKNIU), Trishal, Mymensingh, has been working under my direct supervision as a **Research Assistant** since July 2025, in the Department of Local Government and Urban Development, JKKNIU.

During this period, he has been engaged in the following ongoing research projects under my supervision:

1. **Trends in Environmental Design and Crime Deterrence: A GIS-Based Analysis of Mymensingh** — applying GIS-based spatial analysis to examine the relationship between urban environmental design and crime deterrence patterns.
2. **Spatiotemporal Dynamics and Predictive Modeling of Traffic Accidents Across the Bangladesh Road Hierarchy** — developing a predictive geospatial framework using GIS and Machine Learning approaches (Random Forest, XGBoost, CNN-LSTM) to analyze traffic accident patterns across the national road network.
3. **A Comparative Analysis of Clinical Waste Management Systems: A Mixed-Methods and Machine Learning Approach** — assessing clinical waste management efficiency and environmental risk using GIS spatial analysis and machine learning techniques.

Throughout his tenure, Mahfuz has demonstrated exceptional discipline and commitment, consistently prioritizing and completing his assigned work with diligence. He has shown a genuine and sustained enthusiasm for the field of Urban Development and Planning, and has distinguished himself as one of the most capable researchers in his session. In coursework and group projects alike, he has repeatedly taken on leadership roles, guiding his teams effectively while maintaining strong respect and consideration for his peers, often accommodating others' inconveniences without complaint.

He is highly proficient in quantitative analysis methods, including R programming, QGIS, ArcGIS Pro, Python, and foundational Machine Learning techniques, as well as qualitative data analysis using NVivo. He is a flexible, adaptable individual with a genuine and continuing thirst for learning and knowledge.

I wish him continued success in his future academic and professional endeavors.

Sincerely,



27.07.2026

Uswatun Mahera Khushi

Assistant Professor

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